SNAP Research: Literature Review & Conceptual Foundations

1. Overview

This review examines the foundations of SNAP Research across three main domains: large language model (LLM) capabilities, prompt engineering techniques, and human-AI interaction design. The goal is to highlight how established research informs the system's architecture and user experience.

2. Foundational Research & Model Selection

GPT Series & In-Context Learning

- Brown et al. (2020) demonstrated that scale alone enables LLMs to perform few-shot and zero-shot tasks effectively without needing specialized fine-tuning.
- Takeaway for SNAP: Structured research assistance can be achieved through precise prompting alone, avoiding the need for expensive custom-trained models.
- Gemini Model Family & Efficiency
- The introduction of Google's Gemini suite highlighted significant advancements in multimodal capabilities and complex reasoning. Specifically, smaller variants like Gemini Flash strike a balance between high-speed inference and low latency.

3. Prompt Engineering Strategies

Chain-of-Thought Reasoning

- Building on work by Wei et al. (2022), requiring models to step through their reasoning improves output accuracy on multi-step tasks.
- Implementation: SNAP embeds mode-specific instructions that guide the model to break down complex queries before providing final answers.
- Structured JSON Formatting

4. Comparative Analysis of Existing Systems

Elicit (2021)

- Focus: Extracts data points and synthesizes findings from scientific literature.

Key Difference: While Elicit is tailored strictly for academic papers, SNAP provides a broader foundation for general-purpose research across diverse topics.

Perplexity AI (2022)

- Focus: Merges live web search with brief AI-generated summaries.

Key Difference: Perplexity prioritizes search indexing, whereas SNAP focuses on guided depth modes (Exploration, Analysis, Deep-Dives) to control research workflows.

Semantic Scholar (2015–Present)

- Focus: Uses classic NLP for paper recommendation graphs and entity extraction.

Key Difference: SNAP relies on modern LLM reasoning to dynamically pull out concepts and suggest follow-up questions in real time.

Author / Source	Year	Primary Focus	Application in SNAP
Brown et al.	2020	In-context few-shot learning	Directs system prompt structure
Wei et al.	2022	Step-by-step reasoning steps	Enhances output quality and depth
Google DeepMind	2024	Gemini architecture performance	Informs core model selection
Liang et al.	2022	HELM evaluation benchmarks	Standardizes testing criteria
Amershi et al.	2019	Human-AI design principles	Shapes UI interaction flow
Nielsen	1994	Usability engineering guidelines	Directs general UX decisions

7. References

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